

# SHUBHAM KALE

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## EDUCATION

### Johns Hopkins University, Whiting School of Engineering

B.S. Applied Mathematics & Statistics, B.A. Economics | Expected 05/29 | GPA: 3.70/4.00

**Honors:** Hodson Trust Scholar (~20 of 45,000 applicants), 1For2 Foundation Scholar (~5 of 1,000); ~\$300K in total scholarships awarded

## TECHNICAL SKILLS

**Languages:** TypeScript/JavaScript, Python, C/C++, Swift/SwiftUI, SQL, R, Java

**Frameworks & Infra:** React, Node, Express, Vite, Flask, FastAPI, REST APIs, PostgreSQL, SQLite, Supabase (Edge Functions), AWS, Google Cloud

**AI & Agents:** LLM application development, agent orchestration and state machines, tool calling, retrieval-augmented generation (RAG), embeddings and vector search (Pinecone), Neo4j, PyTorch

**Workflow:** Git & GitHub (branching, pull requests, code review), CI pipelines, Playwright, testing, technical documentation

## ENGINEERING EXPERIENCE

### Summer Atlantic Capital LLC & SAC Nexus

Jun 2026 – Present

Product Development Team Member — Data Scientist

- Own the ML core of a security and compliance anomaly-detection system, shipped as a versioned **FastAPI microservice**; built the repository and **CI scaffold**, event-ingestion pipeline, and detect\_anomalies API
- Built a temporal sequence model (RNN/Transformer) over high-volume event streams and a **learning-to-rank** alert-prioritization model, cutting manual analyst review load
- Engineered a pluggable **LLM explanation layer with retrieval-augmented generation**; documented assumptions, limitations, and performance metrics to institutional standards

### Open Health Intelligence (ClarityAtlas) | Baltimore, MD

Sep 2025 – Present

Co-Founder & Technical Lead

- Built and **shipped a full-stack iOS application end to end** (Swift, SwiftUI, Supabase backend), now running in a **live pilot at Johns Hopkins**
- Architected the entire backend solo: **HealthKit integration** (Core Motion, step count, gait speed), a geofenced behavioral-prompt system, and a “Sleep Twins” matching service on **Supabase Edge Functions** using weighted Euclidean distance
- Designed the **passive sleep-estimation algorithm** inferring sleep windows from phone-interaction and motion signals; received a non-binding ~\$75K investment indication and presented to JHU President Ron Daniels and DSAI leadership

### SnapSort | Phoenix, AZ

Jan 2024 – Aug 2025

Founder & Sole Engineer

- Built and shipped an AI photo-sorting web app solo from zero to **300+ users** (Flask, AWS, multi-model API pipeline); raised ~\$5K in non-dilutive funding and led a team of 4
- Trained a **convolutional neural network in PyTorch** for facial-landmark prediction (GridSearchCV for tuning and model selection), reaching ~95% PCK

### Intelehealth | Hybrid (Baltimore & India)

Sep 2025 – Jun 2026

Software Engineering / AI Research Intern

- Built a resource-availability scoring pipeline in Python (facility-tier classification, Google Maps + haversine distance across 32 villages, weather API integration) for a rural telemedicine platform

## PROJECTS

### [tSearch — Talent Discovery Engine](#) | TypeScript, Neo4j

2026

Sole Developer — built for Cory Levy (Founder, Z Fellows)

- Shipped an **MVP in four days** for a problem scoped by the founder of Z Fellows; built a pipeline resolving identity, expanding a collaboration graph via the **GitHub API**, and rendering a radial graph UI (Express + Vite)
- Added a **rubric-driven LLM judge layer** running in parallel per dimension, producing evidence-cited scores and an emailed digest; iterated through two rounds of feedback into an introduction to his team

### [Gloria — AI Memory Companion](#) | React/TypeScript, Python/Flask

2025

Sole Developer

- Built a **full-stack LLM product end to end** for three personas: React/TypeScript frontend, Flask REST API, OpenAI embeddings, Pinecone vector search, and a Neo4j graph driving an agent state machine

### Dispatch — Workflow Automation Agent | TypeScript, Playwright, SQLite

2026

Sole Developer

- Built a deterministic **browser-automation agent** with a **SQLite-backed state machine** as single source of truth, **fail-closed feature flags** on every mutating capability, and a four-stage validation ladder requiring proof against live pages, not fixtures

## PUBLICATIONS & AWARDS

- Sole author, *Journal of Student Research* (2024); co-author, *Microbiology Resource Announcements* (ASM, 2024); 63rd National JSHS Finalist (top ~150 of 8,000+)